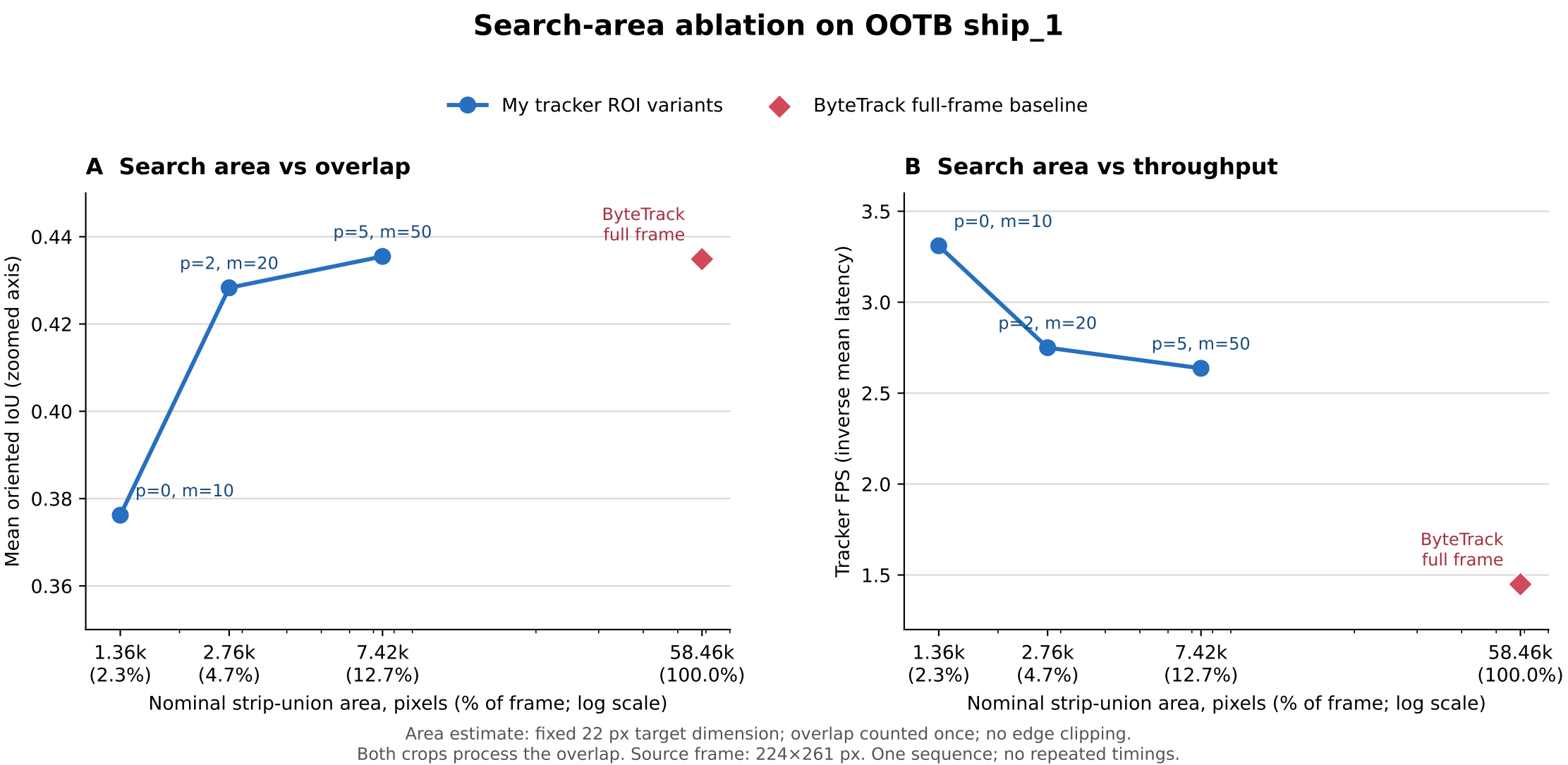
Search-area ablation on OOTB ship_1
My tracker ROI variants
ByteTrack full-frame baseline
A Search area vs overlap
Mean oriented IoU (zoomed axis)
0.44
0.42
0.40
0.38
0.36
p=0, m=10
p=2, m=20
p=5, m=50
ByteTrack full frame
1.36k (2.3%)
2.76k (4.7%)
7.42k (12.7%)
58.46k (100.0%)
Nominal strip-union area, pixels (% of frame; log scale)
B Search area vs throughput
Tracker FPS (inverse mean latency)
3.5
3.0
2.5
2.0
1.5
p=0, m=10
p=2, m=20
p=5, m=50
ByteTrack full frame
1.36k (2.3%)
2.76k (4.7%)
7.42k (12.7%)
58.46k (100.0%)
Nominal strip-union area, pixels (% of frame; log scale)
Area estimate: fixed 22 px target dimension; overlap counted once; no edge clipping.
Both crops process the overlap. Source frame: 224×261 px. One sequence; no repeated timings.